TriNetra (Third Eye) - Offline Evaluation Guide

Complete Step-by-Step Manual for Proving 100% Autonomous Edge AI Navigation

ARCHITECTURE: PWA + WEBGL TENSORFLOW.JS | HARDWARE: LOCAL SMARTPHONE SENSORS

1. Executive Summary & The Offline Imperative

For a visually impaired person, navigation safety cannot depend on an internet connection. Blind pedestrians routinely enter underground subway stations, basement elevators, steel-reinforced building corridors, and rural transit routes where cellular connectivity completely drops.

If an assistive mobility aid requires transmitting camera frames to a cloud server to calculate distances, the user is rendered electronically blind and vulnerable the instant signal is lost. TriNetra solves this through a Dual-Layer Perception Architecture: all safety-critical collision avoidance runs 100% on-device.

[!] THE TRINETRA ARCHITECTURAL GUARANTEE

Physical safety, 3D spatial acoustic sonar, looming collision alerts, target object seeking, and tactile full-screen gestures operate completely on the phone GPU and CPU. Zero cloud requests are made during core walking navigation.

2. Rule 0: One-Time Warmup & Service Worker Caching

Before taking the phone into an offline demonstration, execute a single pre-caching cycle:

* Step A: Open TriNetra once in Google Chrome or Safari with internet active.
* Step B: The PWA Service Worker (Workbox) caches all HTML, CSS, JavaScript, and the 2.5 MB TensorFlow.js MobileNetV2 neural network weights directly into local browser storage.
* Step C: (Recommended) Tap browser settings and select "Add to Home Screen" to install it as an app.
* Result: Once cached, TriNetra never needs internet again for core collision avoidance.

3. Live Airplane Mode Demonstration (Step-by-Step)

Perform this exact sequence in front of judges, evaluators, or test audiences:

Step 1: Enter 100% Airplane Mode in View of Judges

Swipe down the phone notification tray. Turn on Airplane Mode. Explicitly disable Wi-Fi and Mobile Data. Show the active airplane icon to the evaluators to prove zero connectivity.

Step 2: Relaunch or Refresh TriNetra

Relaunch the app from the home screen. Notice that no "No Internet" dinosaur appears; the app boots instantly in under 1 second from the local Service Worker cache.

Step 3: Tap Display to Unlock Audio

Tap anywhere on the glass. The device on-board Text-to-Speech engine announces:
>> "TriNetra active. Scanning started." (Spoken locally via Web Speech API).

Step 4: Live 3D Spatial Sonar Avoidance

Hold a chair, bag, or hand 1 to 2 meters in front of the camera. The on-device TF.js model detects the object with sub-100ms latency, emitting stereo pings that pan left/right and scale pitch (440Hz -> 880Hz).

Step 5: Target Finder (Seek Object)

Two-Finger Swipe Right. App announces: "Target set to chair. Point camera to locate." Pan the phone across the room. When centered on a chair, hear the rapid 988 Hz acoustic beacon tone.

Step 6: Light Detector Tone

Two-Finger Swipe Down. Aim at a light fixture or window. Audio pitch rises continuously with brightness.

Step 7: Hardware Flashlight & Telemetry

Swipe Up to toggle camera LED torch. Swipe Down to hear: "Full audio mode. Running at 4 FPS. Path clear."

Step 8: Emergency Kill-Switch

Three-Finger Tap. Instantly halts all audio loops: "TriNetra emergency pause. Tap screen to resume."

4. Comprehensive Feature Matrix: Offline vs. Online

Clear breakdown of which subsystems operate locally versus those utilizing cloud GenAI:

Feature / Tool	Offline Status	Underlying Engine & Local Mechanism
Obstacle Detection	100% OFFLINE	TensorFlow.js + MobileNet-V2 on phone WebGL GPU.
3D Binaural Sonar	100% OFFLINE	Web Audio API PannerNode and oscillators on phone DSP.
Looming Collision Alert	100% OFFLINE	Bounding box area expansion rate: delta-area/delta-t > 15%/s.
Target Finder (Seek)	100% OFFLINE	Local COCO-SSD class filtering + acoustic directional beacon.
Light Tone Detector	100% OFFLINE	HTML5 canvas pixel luminance sampling to oscillator pitch.
LED Flashlight (Torch)	100% OFFLINE	MediaStreamTrack applyConstraints torch control.
Tactile Gestures	100% OFFLINE	Centroid multi-touch tracking + Web Vibration API haptics.
Audio Mode Switching	100% OFFLINE	Full, Voice-Only, Sonar-Only, Emergency-Only local state.
Emergency Kill-Switch	100% OFFLINE	Three-finger tap instant oscillator and inference cancel.
Gemini Scene Narrator	CLOUD REQUIRED	Google Gemini 2.5 Flash for natural paragraph descriptions.
Verbatim OCR Reader	CLOUD REQUIRED	High-res 1600px frame to Gemini Vision for document reading.
Banknote Reader	CLOUD REQUIRED	Gemini Vision fine denomination and currency verification.

5. Architectural Isolation Test (Cloud Fallback)

An essential mark of engineering rigor is graceful degradation. In Airplane Mode, perform a Two-Finger Tap (Gemini Scene Description) or Two-Finger Double Tap (Banknote Reader).

[!] EVALUATOR OBSERVATION POINT

The app attempts the API fetch, catches the offline state, and calmly speaks: "Scene description failed. Please check internet connection."

Key Proof for Judges: The obstacle collision avoidance radar, audio sonar pings, and safety alerts continue running in the background without freezing, crashing, or stuttering.

6. The 60-Second Evaluator Presentation Script

[STEP 1: Hold up smartphone showing Airplane Mode active]

"Judges, imagine a visually impaired student entering an underground metro station, an elevator, or a rural road where cellular reception drops completely. If an assistive walking aid relies on cloud servers, the user is left completely stranded and blind. That is why TriNetra was engineered with an offline-first Edge AI architecture."

[STEP 2: Relaunch TriNetra and tap the screen to activate]

"Notice that my phone is in 100% Airplane Mode - zero Wi-Fi, zero mobile data. When I launch TriNetra, our Service Worker boots the entire app from local cache in less than one second. The camera stream, the TensorFlow.js neural network running on the phone GPU, our 3D binaural spatial sonar, and our tactile gesture surface are executing 100% on this smartphone."

[STEP 3: Walk toward a chair or hold a hand 1 meter in front of camera]

"As an obstacle enters the frame, the user hears sub-100 millisecond directional audio pings in their ears, warning them of collision hazards with zero server latency, complete privacy, and zero data cost. Their fundamental walking safety never depends on a cell tower."